Currency Exchange — Provider Recommendation

Production decision note for Lead / Project Manager · POC: Angular + Django

1. Context

We need server-side FX conversion for amounts entered in INR (or local currency) into USD, AED, SAR, and others, using the rate for a user-selected date. Keys must stay on the Django backend; each transaction should persist **rate**, **rate_date**, and **provider** for audit.

2. Reference baseline (from comparison pack)

Date: **22/10/2026** · Amount: **10,000 INR**

Reference	USD	AED
Original / target mid-market	105.43	388.90

3. Services researched

Three services were evaluated during the POC:

1. **Open Exchange Rates** — https://openexchangerates.org/
2. **ExchangeRate-API** — https://www.exchangerate-api.com/
3. **CurrencyAPI / community CDN feeds** — prototype only, not for production

4. Recommendation

Primary (production): Open Exchange Rates (paid plan).
If price is the main constraint: ExchangeRate-API (paid plan) — easy to switch via env (`FX_PROVIDER=exchangerateapi`).

5. Why Open Exchange Rates over ExchangeRate-API

- **Historical / date-based rates are first-class on commercial plans** — critical when invoices or bookings must use the FX of the user-entered date, not only “today’s” mid-market.
- **Stronger production posture** — App ID model, widely adopted in backend apps, clearer commercial licensing and ops maturity for a company product.
- **Fits multi-currency pipelines** — USD-base industry pattern with cross-rates (INR, AED, SAR, etc.) implemented once in Django; easier to standardize across environments.
- **Audit-friendly** — dated historical endpoints align with storing rate + rate_date per transaction.
- **Trade-off acknowledged:** ExchangeRate-API often looked slightly closer on AED in our free-tier spot checks and is typically a bit cheaper. We still prefer OXR for production capability; ER-API remains the budget switch.

6. Pros / cons (paid intent)

	Open Exchange Rates	ExchangeRate-API
Best for	Production + historical dates	Cost + simple latest/pair FX
Pros	Historical, mature backend API, audit-friendly dated rates	Good USD/AED feel in POC, simpler/cheaper paid path

Cons	Slightly higher price; AED may need bank cross-check	Weaker historical story on lower tiers; less “enterprise FX” standard
Free ballpark	~1,000 req/month (mainly latest)	~1,500 req/month (latest)

7. Implementation status

Code default is now **Open Exchange Rates** (`FX_PROVIDER=openexchangerates`). ExchangeRate-API remains wired for a one-line env switch. Community CDN is retained only as a last-resort prototype fallback.

8. Compliance note

For statutory / tax / books in India, confirm whether **RBI or bank mid-rates** are mandatory. Retail FX APIs are appropriate for in-app conversion when we persist the exact rate used; they may not replace official reference rates for accounting.

Decision: Adopt Open Exchange Rates (paid) as primary. Keep ExchangeRate-API as price-sensitive alternate.